



AMITY UNIVERSITY
— **UTTAR PRADESH** —

DOMAIN: ENGINEERING AND TECHNOLOGY

INTERNSHIP-I

i. Student Details

Name	Samay Kumar
Enrollment Number	A2305325001
Programme	B.Tech – Information Technology
Year / Semester	2nd Year / 3rd Semester
Academic Session	2026–2027
Section	B.Tech IT
Institution	Amity School of Engineering & Technology, AUUP, Noida

ii. Faculty Guide

Name	Ms. Veenu Singh
Designation	Faculty Guide, ASET – Amity University, Noida

iii. Topic

Centralized vs Peer-to-Peer Open-Source VPN Architectures in Linux Environments

iv. Objective(s)

- To analyze and compare centralized VPN architectures (WireGuard) and peer-to-peer mesh VPN architectures (Headscale) deployed on Linux systems.
- To evaluate performance across key metrics: latency, throughput, CPU utilization, and packet loss using standard Linux benchmarking tools.
- To assess fault tolerance and resilience of each VPN architecture under simulated node and server failure conditions.
- To examine the security models, encryption mechanisms, trust assumptions, and infrastructure dependencies of each VPN solution.
- To design and implement a real-time performance visualization dashboard displaying live benchmark data, topology diagrams, and node health status.
- To produce a structured comparative analysis to inform practical VPN deployment decisions in Linux-based infrastructure.